CMSE 801 - Day 21

Random numbers

Nov 21, 2022

General Course Announcements

- Homework #5 is due Wednesday Nov 30 at 11:59 p.m.
- Project's deadline Sunday Dec 4 at 11:59 p.m.

Questions/comments from Day 21 PCA

- In the dice case, you can do it with a dice list using random.choice or using random.randint. Both seem to converge differently and seed does not affect random.choice. Why is that?
- going over how to do subplots for the various types of distributions - felt lost once again...
- Should we finish everything in final project before the project review?

In [1]:

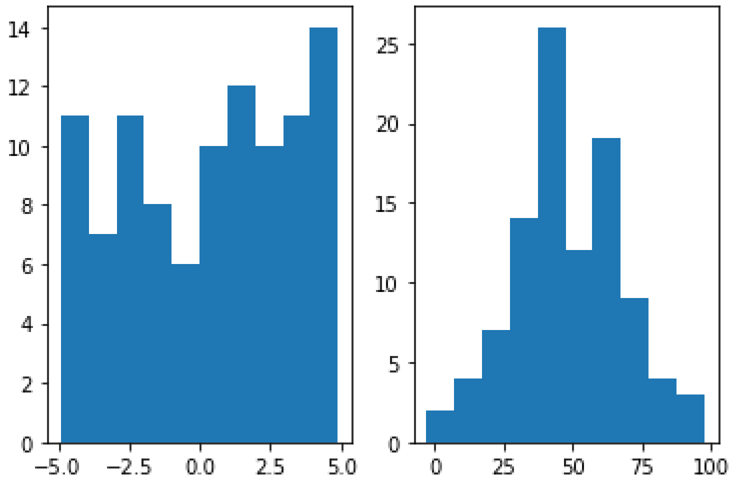
```
import random
import matplotlib.pyplot as plt
import numpy as np
%matplotlib inline
```

In [2]:

```
fig, ax = plt.subplots(1,2)
count = 100
vals = np.random.uniform(-5,5,size=count)
ax[0].hist(vals)
mean = 50
stddev = 20
vals = np.random.normal(mean,stddev,size=count)
ax[1].hist(vals)
```

Out[2]:

```
(array([ 2.,  4.,  7., 14., 26., 12., 19.,  9.,  4.,  3.]),
 array([-2.71167979,  7.32582187, 17.36332353, 27.40082519, 37.43832685,
        47.47582851, 57.51333017, 67.55083183, 77.58833349, 87.62583515,
        97.66333681]),
 <BarContainer object of 10 artists>)
```



In [3]:
```
foo = ['a', 'b', 'c', 'd', 'e']
np.random.choice(foo,size=2,replace=False)
```

Out[3]:
```
array(['d', 'a'], dtype='<U1')
```

In [4]:
```
for count in [100,1000,10000]:
    ones = 0
    sixes = 0
    other = 0
    for i in range(count):
        roll = random.randint(1,6)
        if roll == 1:
            ones += 1
        elif roll == 6:
            sixes += 1
        else:
            other += 1
    fractions = [ones/count,sixes/count,other/count]
    print(fractions)
print("1/6 = ",1/6)
```

```
[0.16, 0.17, 0.67]
[0.141, 0.171, 0.688]
[0.1627, 0.1756, 0.6617]
1/6 =  0.16666666666666666
```

In [5]:
```
random.seed(1)
print(random.randint(1,6))
foo = ['a', 'b', 'c', 'd', 'e']
random.seed(1)
print(np.random.choice(foo))
```

```
2
e
```

In [6]:
```
random.seed(1)
print(random.randint(1,6))
foo = ['a', 'b', 'c', 'd', 'e']
np.random.seed(1)
print(np.random.choice(foo))
```

```
2
d
```